THE UNIVERSITY OF MELBOURNE

Student number

Assignment 9 Semester 1, 2023

School of Mathematics and Statistics

MAST10012 Introduction to Mathematics

Submission deadline: 5:00 PM (17:00, AEST), Thursday 18 May 2023

This assignment consists of 9 pages (including this page)

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Instructions to Students

Writing

- If you have a printer (or can access one), print the assignment one-sided. Write your answers in the boxes provided on the assignment that you have printed.
- If you do not have a printer, and you know how to annotate a PDF using a tablet, then annotate your answers directly onto the assignment PDF and save a copy for submission.
- If you cannot use the template, write your answers on blank A4 paper. Mimic the template by numbering your pages, and note which answers to write on each page.
- Record your student ID number in the top right corner of this cover page (or the first page of your submission), and your details at the bottom of this page (or the first page).
- Provide complete and fully justified solutions for each question.
- Write legibly, and use appropriate notation. Marks will not be awarded to unclear work.
- If you run out of space for a question, append additional work to the end of your submission. Make a note ("See page X") in the relevant question box warning the marker.
- Teaching staff cannot help you directly with assignment questions. Please ask on the Ed Discussion board if you need clarification on the wording of a question.

Scanning

- Put the pages in number order and the correct way up. Add any extra pages to the end. Use a scanning app to scan all pages to PDF. Do not scan on an angle. Crop images to the page borders. Make sure that your PDF file is readable. Material that cannot be read will not be marked.

Submitting

- Go to the Gradescope window. Choose the relevant Canvas assignment. Make sure that you upload the correct PDF file. You should get a Gradescope confirmation email.
- Assignments submitted late will incur a penalty of 4% (of the total mark available) for every hour or part thereof up to 24 hours. Written assignments cannot be submitted more than one day (24 hours) late.

Student Details

- You must provide your student identification (ID) number in the box at the top of this page.
- Do not record your name and tutorial details on this paper. Only your student identification (ID) number is required.

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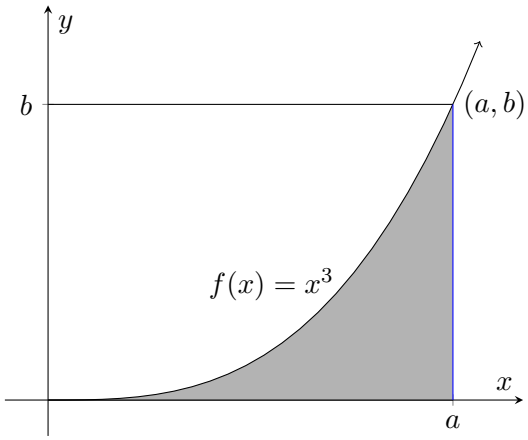
Question 1

In the figure below, B is a point on the curve $f(x) = x^3$. Prove that the shaded region is one quarter of the area in the rectangle with bottom left corner at $(0, 0)$ and top right corner touching the curve at a point (a, b).
Hint: consider the value of the y-coordinate, b, in the point.



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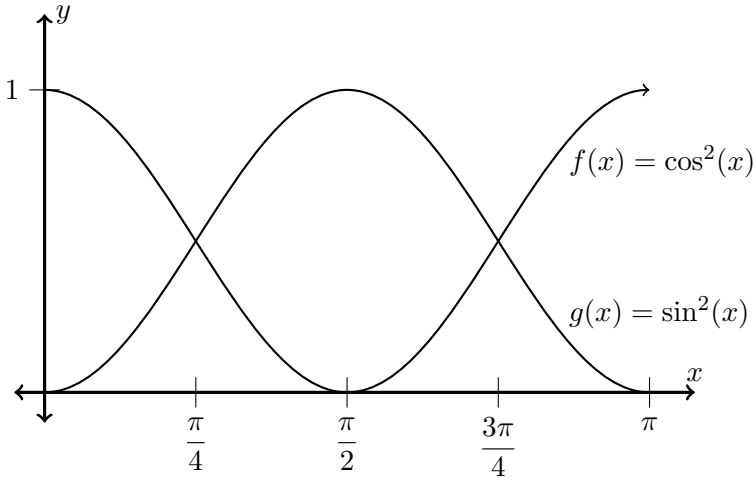
Question 2

Consider the two functions $f(x) = \cos^2(x)$ and $g(x) = \sin^2(x)$ plotted together on the graph:



(a) Show that the functions $f(x)$ and $g(x)$ intersect at $x = \frac{\pi}{4}$ and at $x = \frac{3\pi}{4}$.

(b) Find the derivative, $w'(x)$, for $w(x) = \sin(x)\cos(x)$.

(c) Use your answers from parts (a) and (b) to find the area enclosed between $f(x) = \cos^2(x)$ and $g(x) = \sin^2(x)$ from $x = 0$ to $x = \frac{3\pi}{4}$.

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Question 3

Find $f(x)$ given that $f'(x) = \dfrac{3}{4x}$, and the function passes through the point $\left(-e^4, 6\right)$.

End of Assignment

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Formula Sheet

Combinatorics

- $n! = n \cdot (n-1) \cdot (n-1) \cdots 3 \cdot 2 \cdot 1$
- $\binom{n}{k} = \dfrac{n!}{(n-k)! \cdot k!}$

Functions

- Lines: $y = mx + c$ Point-slope formula: $y = m(x-a) + b$
- Quadratics: $y = ax^2 + bx + c = \underbrace{a(x-r)(x-s)}_{\text{Factorised}} = \underbrace{a(x-h)^2 + k}_{\text{Turning point form}}$, use: $\frac{b}{2a}$
- Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, check: $\Delta = b^2 - 4ac$

Exponentials

- $a^0 = 1$
- $a^1 = a$
- $a^{-1} = \dfrac{1}{a}$
- $a^{\frac{1}{2}} = \sqrt{a}$
- $a^x \cdot a^y = a^{x+y}$
- $\dfrac{a^x}{a^y} = a^{x-y}$
- $(a^x)^y = a^{x \cdot y}$
- $\dfrac{a^x}{b^x} = \left(\dfrac{a}{b}\right)^x$
- $a^x \cdot b^x = (a \cdot b)^x$
- $a^{\frac{x}{y}} = \sqrt[y]{a^x}$
- Change of base: $a^x = c^{x \log_c(a)}$ for any $c \in \mathbb{R}$

Logarithms

- $\log_e(x) = \log(x) = \ln(x)$
- $\log_a(1) = 0$
- $\log_a(a) = 1$
- $\log_a(p^x) = x \cdot \log_a(p)$
- $\log_a(p \cdot q) = \log_a(p) + \log_a(q)$
- $\log_a\left(\dfrac{p}{q}\right) = \log_a(p) - \log_a(q)$
- Change of base: $\log_a(p) = \dfrac{\log_c(p)}{\log_c(a)}$ for any $c \in \mathbb{R}$

Trigonometric Identities

- $\sin(x + 2\pi) = \sin(x)$
- $\cos(x + 2\pi) = \cos(x)$
- $\sin(\pi - x) = \sin(x)$
- $\cos(\pi - x) = -\cos(x)$
- $\sin(-x) = -\sin(x)$
- $\cos(-x) = \cos(x)$
- $\sin(x + y) = \sin(x)\cos(y) + \cos(x)\sin(y)$
- $\cos(x + y) = \cos(x)\cos(y) - \sin(x)\sin(y)$
- $\sin(2x) = 2\sin(x)\cos(x)$
- $\cos(2x) = \cos^2(x) - \sin^2(x)$
- $\sin^2(x) = \frac{1-\cos(2x)}{2}$
- $\cos^2(x) = \frac{1+\cos(2x)}{2}$

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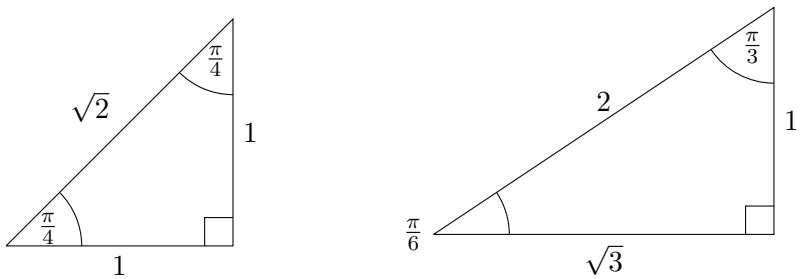
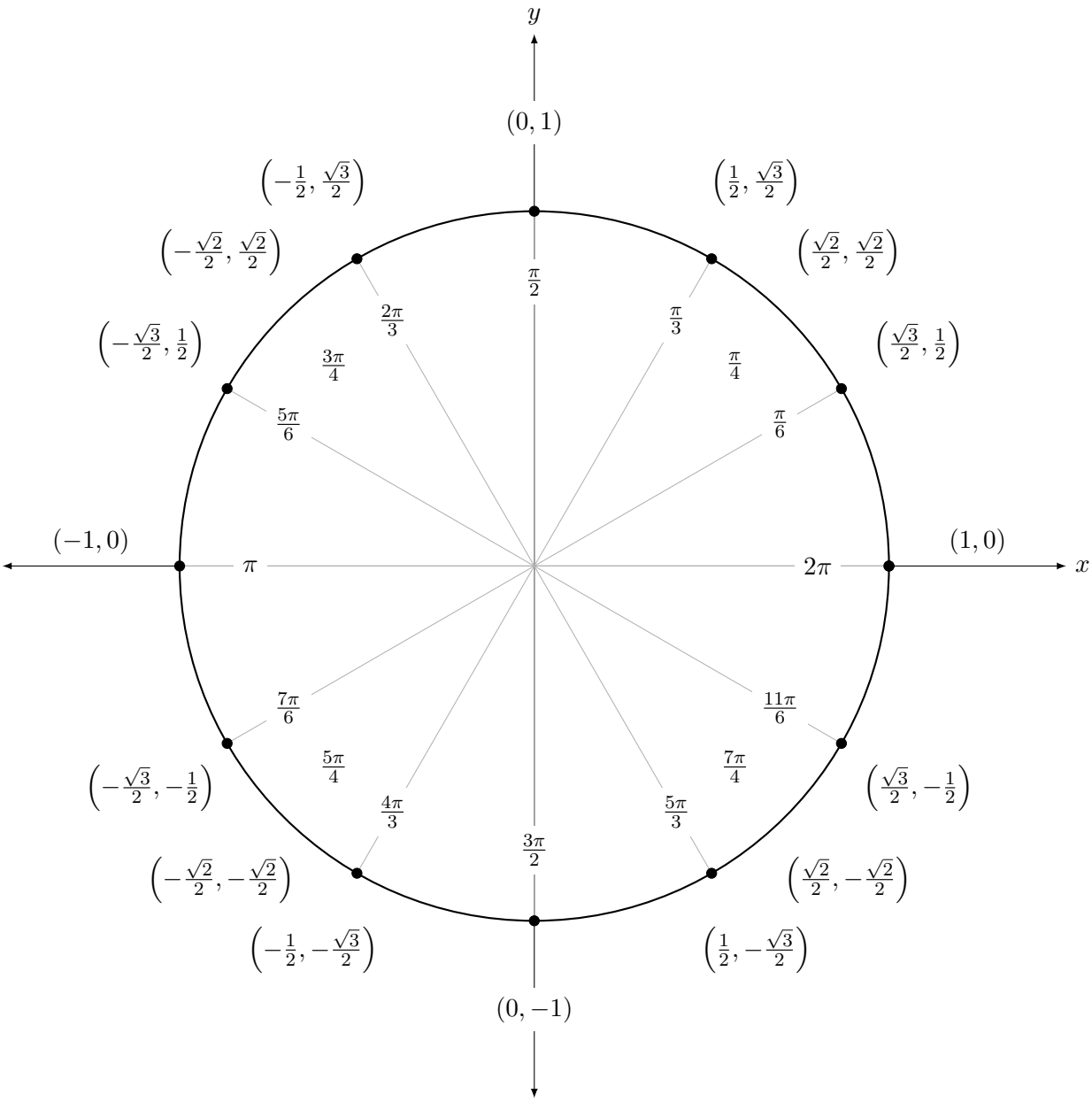
Reference Triangles



Unit Circle



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Limits

If we have $\lim_{x\to a} f(x) = L$, $\lim_{x\to a} g(x) = M$ and $c \in \mathbb{R}$ is a constant, then:

- $\lim_{x\to a} (f(x) + g(x)) = L + M$
- $\lim_{x\to a} (f(x) \cdot g(x)) = L \cdot M$
- $\lim_{x\to a} (f(x))^n = L^n$
- $\lim_{x\to a} c \cdot f(x) = c \cdot L$
- $\lim_{x\to a} c = c$
- $\lim_{x\to a} x^n = a^n$
- $\lim_{x\to a} (f(x) - g(x)) = L - M$
- $\lim_{x\to a} \left(\frac{f(x)}{g(x)}\right) = \frac{L}{M}$
- $\lim_{x\to a} (f(x))^{\frac{1}{n}} = \sqrt[n]{L}$
- $\lim_{x\to a} x = a$
- $\lim_{x\to a} x^{1/n} = \sqrt[n]{a}$

Calculus

- $f'(x) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$
- $\frac{d}{dx}(c) = 0$
- $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$
- $\frac{d}{dx}(\log(x)) = \frac{1}{x}$
- $\frac{d}{dx}(\cos(x)) = -\sin(x)$
- $\frac{d}{dx}(x^n) = n \cdot x^{n-1}$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\sin(x)) = \cos(x)$
- $\frac{d}{dx}(\tan(x)) = \frac{1}{\cos^2(x)}$
- Product rule: $(f \cdot g)'(x) = \frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
- Quotient rule: $\left(\frac{f}{g}\right)'(x) = \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$
- Chain rule: $(f \circ u)'(x) = \frac{d}{dx}(f(u(x))) = f'(u(x)) \cdot u'(x)$

Geometry

- For a triangle with base-length b and height h, the area is $\frac{1}{2}bh$.
- For a circle with radius r, the perimeter is $2\pi r$ and the area is πr^2.
- For a cylinder with radius r and height h, the volume is $\pi r^2 h$.
- For a pyramid or cone with base area A and height h, the volume is $\frac{1}{3}Ah$.
- For a sphere with radius r, the volume is $\frac{4}{3}\pi r^3$.

Probability

- $Pr(X \leq x) = 1 - Pr(X > x)$
- $Pr(B|A) = \frac{Pr(A \cap B)}{Pr(A)}, \quad Pr(A) \neq 0$
- For a discrete random variable X:
  $Pr(X \leq x) = \sum_{k=1}^{x} Pr(X = k)$

Binomial Distribution

For $X \stackrel{d}{=} \text{Bin}(n, p)$:

- $Pr(X = k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$
- $\mathbb{E}(X) = n \cdot p$
- $\text{Var}(X) = n \cdot p \cdot (1-p)$

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The Standard Normal Distribution Table

To standardise the normally distributed random variable X with population mean μ_X and population standard deviation σ_X, use:

$$X \stackrel{d}{=} N\left(\mu_X, \sigma_X^2\right) \quad \Rightarrow \quad Z = \frac{X-\mu}{\sigma} \stackrel{d}{=} N\left(0, 1^2\right)$$

For the standard normal distribution, $Z \stackrel{d}{=} N\left(\mu_Z = 0, \sigma_Z^2 = 1^2\right)$, the probability $Pr(Z \leq z) = Pr(Z < z)$ for a specified z is given to four decimal places in the below table. Note that z is specified to three decimal places, indicated in the row and column headers.

z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999